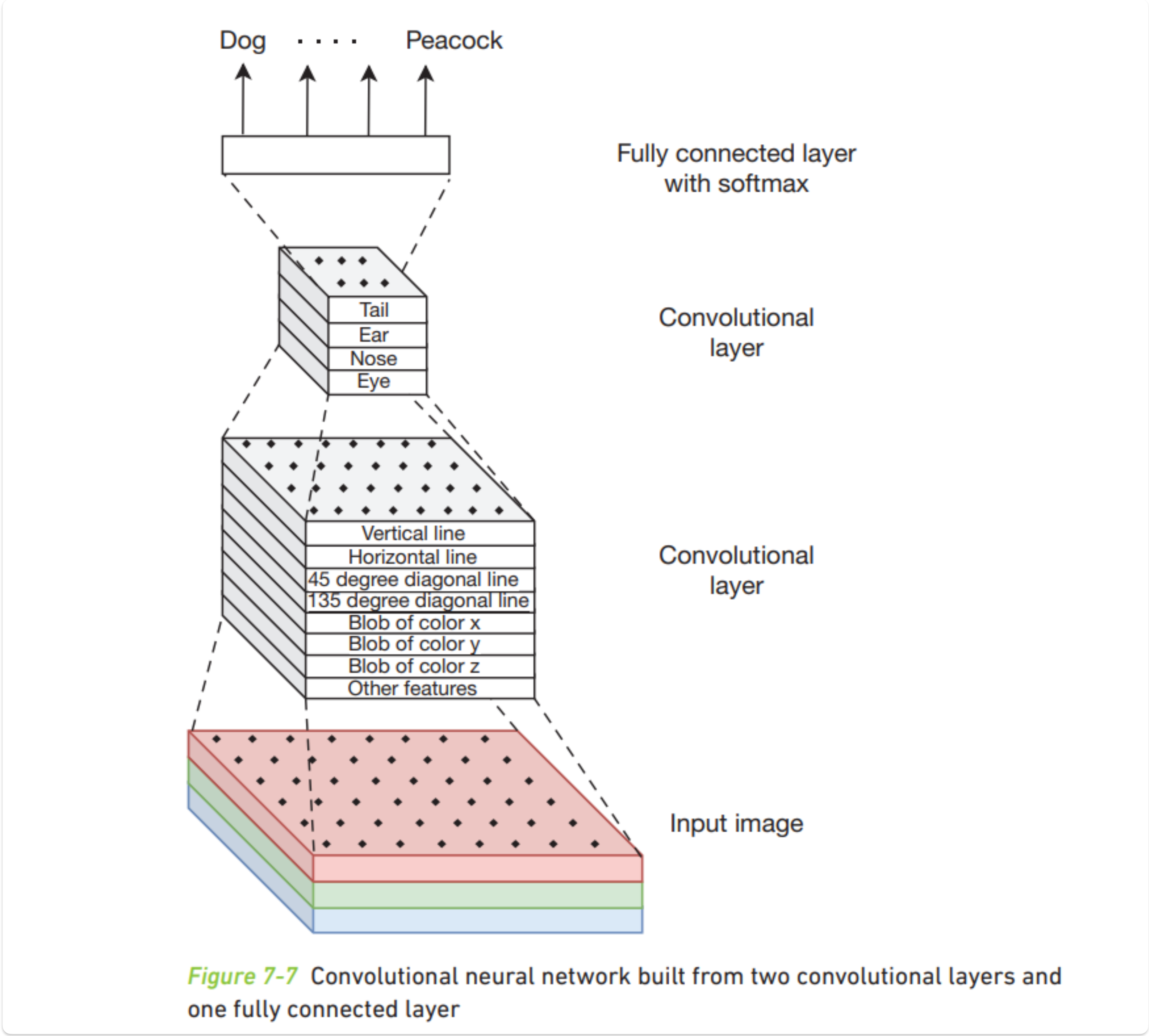


4. The Necessity of Gradually Decreasing Resolution:

A. as shown in the Figure 7-7, the resolution ($\equiv$ num of neurons per feature map) of the first convolutional layer is lower than the resolution of the image, furthermore, the second convolutional layer is lower than the resolution of the first convolutional layer.



B. the reduction in resolution can be achieved either by using a stride greater than 1 or by applying max pooling.

C. because of the hierarchical way neurons are connected, a single neuron in the higher convolutional layer is influenced by a larger pixels from the image than a single neuron in the lower convolutional layer. consequently, the receptive field of a single neuron in the higher convolutional layer is bigger than the receptive field of a single neuron in the lower convolutional layer, even if both of them share the same convolutional kernel size. this arrangement enables neurons in the higher convolutional layer to detect more complex features.

D. the bottom convolutional layer:

- $\Rightarrow$ kernel size = $3 \times 3 \times 1$, which can be simplified into 3.
- $\Rightarrow$ stride = 2 ↪ due to only one feature map in this layer
- $\Rightarrow$ receptive field = each neuron receives 3 pixels, resulting in a receptive field of 3 pixels.
- $\Rightarrow$ grid coverage = requires 4 neurons to maintain the grid in this convolutional layer and ensure all pixels in the image are fully covered.

the top convolutional layer:

- $\Rightarrow$ kernel size = $2 \times 2 \times 1$, which can be simplified into 2.
- $\Rightarrow$ stride = 1 ↪ due to only one feature map in this layer
- $\Rightarrow$ receptive field = each neuron receives 2 computed neurons from the preceding layer, through the hierarchical connectivity, Neuron A's receptive field expands directly to 4 pixels.
- $\Rightarrow$ grid coverage = requires only 2 neurons to maintain the grid in this convolutional layer and ensure all pixels in the image are fully covered, which effectively decreases resolution in this convolutional layer.

